POLYNOMIAL EGYPTIAN SUMS

GPT-5.5 PRO

Abstract. Let $p \in \mathbb{Q}[x]$ be integer-valued on $\mathbb{Z}$, have positive leading coefficient, and have no fixed divisor on the positive integers. We prove that, for each $\alpha \in \mathbb{Q}_{>0}$, all sufficiently large integers m can be written as $\sum_i p(n_i)$, where the n_i are distinct arbitrarily large positive integers satisfying $\sum_i 1/n_i = \alpha$. Consequently, if $p \in \mathbb{Q}[x]$ is zero or has positive leading coefficient, then $\{p(n) + 1/n : n \in \mathbb{N}\}$ is strongly complete. The proof combines the Roth–Szekeres–Graham theorem on complete polynomial sequences with reciprocal-preserving switches and a finite congruence correction.

Throughout, $\mathbb{N} = \{1, 2, \ldots\}$. If $S \subseteq \mathbb{Z}$, we write $\gcd S$ for the non-negative generator of the ideal of $\mathbb{Z}$ generated by S. Thus, zeros do not affect the gcd unless all elements are zero.

For a sequence of integers $(s_1, s_2, \ldots)$, write

$$\mathrm{FS}(s_1, s_2, \ldots) = \left\{ \sum_{i \in I} s_i : I \subseteq \mathbb{N} \text{ finite} \right\}.$$

Terms with equal numerical value but different indices are counted as distinct possible summands.

Theorem 1. *Let $\alpha \in \mathbb{Q}_{>0}$, $L \geq 1$, and let $p \in \mathbb{Q}[x]$ satisfy $p(\mathbb{Z}) \subseteq \mathbb{Z}$. Suppose that the leading coefficient of p is positive and that no integer $d \geq 2$ divides $p(n)$ $(\forall n \geq 1)$. Then there is m_0 such that, for all integers $m \geq m_0$, there exist distinct positive integers*

$$L < n_1 < \cdots < n_k$$

with

$$\sum_{i=1}^{k} \frac{1}{n_i} = \alpha, \qquad \sum_{i=1}^{k} p(n_i) = m.$$

Taking $\alpha = 1$ gives Erdős Problem #283, from [3, p. 32]; see also [1]. Corollary 7 addresses Erdős Problem #351 [3, p. 58], [2], with the necessary correction, noted in the discussion of [2], that one should allow $p = 0$ and otherwise assume positive leading coefficient.

We shall use the following theorem of Roth–Szekeres and Graham [5, 4].

Theorem 2 (Roth–Szekeres–Graham). *Let $f \in \mathbb{Q}[x]$ be nonconstant, with positive leading coefficient. Suppose that*

$$f(n) \in \mathbb{Z}_{>0} \qquad (n \geq 1)$$

and

$$\gcd\{f(n) : n \geq 1\} = 1.$$

Then there is X_f such that all integers $X \geq X_f$ belong to

$$\mathrm{FS}(f(1), f(2), f(3), \ldots).$$

Equivalently, all sufficiently large integers are sums of distinct terms from $(f(n))_{n \geq 1}$.

We use Theorem 2 precisely in the fixed-divisor form stated above. This is Graham's complete-polynomial-values theorem, with the Roth–Szekeres theorem as the asymptotic input in its proof.

Date: May 3, 2026.

2020 *Mathematics Subject Classification.* Primary 11D68; Secondary 11B13, 11C08.

Key words and phrases. Egyptian fractions, complete sequences, additive bases, integer-valued polynomials, fixed divisors.

1. Egyptian switches

We first record two elementary facts about Egyptian fractions.

Lemma 3. *Let $R \in \mathbb{Q}_{>0}$ and $L \geq 0$. Then R has a finite expression as a sum of reciprocals of distinct integers greater than L. Moreover, such expressions exist with all sufficiently large numbers of terms.*

Proof. Replacing L by an integer $L' \geq \max\{1, L\}$, it is enough to prove the result with L' in place of L. Hence, we may suppose L is a positive integer. For $N \geq L$, set

$$H_N^{(L)} = \sum_{j=L+1}^{N} \frac{1}{j}.$$

Choose $N \geq L$ maximal with $H_N^{(L)} < R$. Then

$$0 < R - H_N^{(L)} \leq \frac{1}{N+1}.$$

Apply the usual greedy algorithm to $R - H_N^{(L)}$. If $S = a/b \in \mathbb{Q}_{>0}$ is in lowest terms and $0 < S \leq 1$, choose $q = \lceil b/a \rceil$. If $S \neq 1/q$, then

$$S - \frac{1}{q} = \frac{aq - b}{bq},$$

where $0 < aq - b < a$. Hence, the greedy process terminates because the numerator strictly decreases. Also, after choosing $1/q$, the next denominator, if any, is larger than q. Since the first greedy denominator for $R - H_N^{(L)}$ is at least $N + 1$, this gives a distinct expansion not colliding with $L + 1, \ldots, N$.

Now, suppose an expansion contains a largest denominator y. Replacing

$$\frac{1}{y} \quad \text{by} \quad \frac{1}{y+1} + \frac{1}{y(y+1)}$$

preserves the represented rational, preserves distinctness, keeps all denominators greater than L, and increases the number of terms by one, because both new denominators are larger than y. Repeating this operation gives all sufficiently large numbers of terms. □

Call a finite set $E \subseteq \{2, 3, 4, \ldots\}$ an *Egyptian pattern* if

$$\sum_{e \in E} \frac{1}{e} = 1.$$

Lemma 4. *Let $T, M \geq 1$ be integers, and let $\rho \in \mathbb{Z}/M\mathbb{Z}$. There is an Egyptian pattern E such that $T \mid e$ $(\forall e \in E)$ and $|E| \equiv \rho \pmod{M}$.*

Proof. Choose a sufficiently large integer k with $k \equiv \rho \pmod{M}$. By Lemma 3, with lower bound 1 if $T = 1$ and lower bound 0 otherwise, there is a finite set $F \subseteq \mathbb{N}$ such that

$$T = \sum_{f \in F} \frac{1}{f}, \qquad |F| = k.$$

Set $E = \{Tf : f \in F\}$. Then the elements of E are distinct, $T \mid e$ $(\forall e \in E)$, and

$$\sum_{e \in E} \frac{1}{e} = \frac{1}{T} \sum_{f \in F} \frac{1}{f} = 1.$$

Also, $E \subseteq \{2, 3, \ldots\}$: for $T = 1$ this follows from the lower bound on F, while for $T > 1$ it follows from $Tf \geq T \geq 2$. Finally, $|E| = |F| \equiv \rho \pmod{M}$. □

For an Egyptian pattern E, define the switching polynomial $Q_E(x) = \sum_{e\in E} p(ex) - p(x)$. If p has degree $r \geq 1$ and leading coefficient $a > 0$, then

$$Q_E(x) = a\left(\sum_{e\in E} e^r - 1\right) x^r + O(x^{r-1}),$$

so Q_E has positive leading coefficient.

Lemma 5. *Let $p \in \mathbb{Q}[x]$ satisfy $p(\mathbb{Z}) \subseteq \mathbb{Z}$. Let $B \geq 1$ be such that $Bp(x) \in \mathbb{Z}[x]$. If $m \geq 1$ and $x \equiv y \pmod{mB}$, then $p(x) \equiv p(y) \pmod{m}$. In particular, mB is a period of $p(n) \pmod{m}$.*

Proof. Write $R(x) = Bp(x) \in \mathbb{Z}[x]$. Since $x \equiv y \pmod{mB}$, each monomial difference $x^k - y^k$ is divisible by mB. Hence, $R(x) - R(y)$ is divisible by mB. But

$$R(x) - R(y) = B(p(x) - p(y)),$$

and the bracket is an integer because p is integer-valued on $\mathbb{Z}$. Therefore, $m \mid p(x) - p(y)$. □

Lemma 6. *Assume that $p \in \mathbb{Q}[x]$, $p(\mathbb{Z}) \subseteq \mathbb{Z}$, $\deg p \geq 1$, the leading coefficient of p is positive, and $\gcd\{p(n) : n \geq 1\} = 1$. Then there are finitely many Egyptian patterns $E_1, \dots, E_s$ such that*

$$\gcd\{Q_{E_i}(n) : 1 \leq i \leq s,\ n \geq 1\} = 1.$$

Proof. Suppose first that the gcd of all integers $Q_E(n)$, as E ranges over all Egyptian patterns and $n \geq 1$, is not 1. Then some prime ℓ divides all such values.

Choose $B \geq 1$ with $Bp(x) \in \mathbb{Z}[x]$, and put $T_\ell = \ell B$. By Lemma 5, T_ℓ is a period of $p(n) \pmod{\ell}$. By Lemma 4, choose an Egyptian pattern E such that $T_\ell \mid e$ ($\forall e \in E$) and $|E| \equiv 0 \pmod{\ell}$. For $n \geq 1$, we have $en \equiv 0 \pmod{T_\ell}$ ($\forall e \in E$), so $p(en) \equiv p(0) \pmod{\ell}$. Since $\ell \mid Q_E(n)$,

$$0 \equiv Q_E(n) = \sum_{e\in E} p(en) - p(n) \equiv |E|p(0) - p(n) \equiv -p(n) \pmod{\ell}.$$

Thus, $\ell \mid p(n)$ for all $n \geq 1$, contradicting $\gcd\{p(n) : n \geq 1\} = 1$. Therefore, the gcd of all values $Q_E(n)$ is 1.

Finally, the ideal of $\mathbb{Z}$ generated by all these values is the whole of $\mathbb{Z}$. Since ideals in $\mathbb{Z}$ are finitely generated, finitely many individual values already generate the unit ideal. Taking the corresponding finitely many patterns $E_1, \dots, E_s$, the asserted gcd is 1. □

2. Proof of the theorem

Proof of Theorem 1. First suppose p is constant, say $p(n) = c$. The positive leading coefficient condition gives $c > 0$, and the no-fixed-divisor condition gives $c = 1$. By Lemma 3, the rational number α has distinct Egyptian expansions, with all denominators greater than L, with all sufficiently large numbers of terms. If such an expansion has exactly m terms, then the corresponding p-sum is m. This proves the theorem in degree 0.

Assume from now on that $r = \deg p \geq 1$, and let $a > 0$ be the leading coefficient of p.

The main slots. Put

$$P = 36, \qquad u_j = Pj + 1, \qquad D_j = u_j u_{j+1}.$$

Then

$$\frac{1}{D_j} = \frac{1}{P}\left(\frac{1}{u_j} - \frac{1}{u_{j+1}}\right).$$

Let

$$E_0 = \{2, 3, 6\}, \qquad A(x) = Q_{E_0}(x) = p(2x) + p(3x) + p(6x) - p(x),$$

and put $\Theta = 2^r + 3^r + 6^r - 1$. The leading coefficient of A is $a\Theta > 0$. Choose $J \geq 1$ so large that

$$A(D_j) > 0\ (j \geq J), \qquad \frac{1}{Pu_J} < \alpha, \qquad D_J > L, \qquad Pu_{J+1} > L.$$

Define $g = \gcd\{A(D_j) : j \ge J\}$. Set

$$\mathcal{D}(x) = (P(J + x - 1) + 1)(P(J + x) + 1) \in \mathbb{Z}[x],$$

so that $\mathcal{D}(t) = D_{J+t-1}$ for $t \ge 1$, and define

$$q(x) = \frac{A(\mathcal{D}(x))}{g} \in \mathbb{Q}[x].$$

Then $q(t) \in \mathbb{Z}_{>0}$ $(t \ge 1)$ and $\gcd\{q(t) : t \ge 1\} = 1$. Moreover, $\mathcal{D}$ has degree 2 and leading coefficient P^2, while A has degree r and leading coefficient $a\Theta$. Thus, q is a nonconstant polynomial of degree $2r$, with positive leading coefficient

$$\kappa = \frac{a\Theta P^{2r}}{g}.$$

By Theorem 2, choose X_0 such that all integers $X \ge X_0$ belong to $\mathrm{FS}(q(1), q(2), q(3), \ldots)$. Set $M_0 = gX_0$ and $\lambda = aP^{2r}$. Since $\Theta > 1$, choose a real number μ satisfying

$$\lambda < \mu < a\Theta P^{2r} = g\kappa.$$

Put $c = \mu/g$, so that $c < \kappa$.

We claim that there is N_{rep} such that, for all $N \ge N_{\mathrm{rep}}$, any multiple M of g with $M_0 \le M \le \mu N^{2r}$ has the form

$$M = \sum_{j \in S} A(D_j)$$

for some $S \subseteq \{J, J + 1, \ldots, N\}$.

Indeed, let $H = N - J + 1$. Since $q(t)/t^{2r} \to \kappa$ and $(H + 1)/N \to 1$, choose N_{rep} so large that, whenever $N \ge N_{\mathrm{rep}}$ and $t > H$, one has $q(t) > cN^{2r}$. For instance, choose $c < c' < \kappa$, use $q(t) \ge c't^{2r}$ for all large t, and then use $c'(H + 1)^{2r} > cN^{2r}$ for all large N.

Now let M be a multiple of g with $M_0 \le M \le \mu N^{2r}$, and write $X = M/g$. Then $X \ge X_0$, so

$$X = \sum_{t \in W} q(t)$$

for some finite $W \subseteq \mathbb{N}$. Since $X \le cN^{2r}$ and all $q(t)$ are positive, no element of W can exceed H. Hence, $W \subseteq \{1, \ldots, H\}$. Taking $S = \{J + t - 1 : t \in W\}$ gives

$$M = \sum_{t \in W} gq(t) = \sum_{j \in S} A(D_j),$$

as claimed.

Correction slots. Put $\delta = \alpha - \frac{1}{Pu_J} > 0$. If $g = 1$, no correction slots are needed; set

$$R_c = 0, \qquad C_0 = 0, \qquad B_* = 0.$$

The empty subset represents the unique residue class modulo 1.

Assume now that $g > 1$. By Lemma 6, choose Egyptian patterns $E_1, \ldots, E_s$ such that

$$\gcd\{Q_{E_i}(n) : 1 \le i \le s,\ n \ge 1\} = 1.$$

Therefore, the residues $Q_{E_i}(n) \pmod g$ generate the additive group $\mathbb{Z}/g\mathbb{Z}$. Choose finitely many pairs (i_σ, a_σ), $1 \le \sigma \le w$, with $1 \le i_\sigma \le s$ and $a_\sigma \ge 1$, such that

$$\rho_\sigma = Q_{E_{i_\sigma}}(a_\sigma) \pmod g$$

generate $\mathbb{Z}/g\mathbb{Z}$.

Let $B_p \ge 1$ be such that $B_p p(x) \in \mathbb{Z}[x]$, and set $T_g = gB_p$. By Lemma 5, T_g is a period of $p(n) \pmod g$. Consequently, for each fixed pattern E_i, the map $n \mapsto Q_{E_i}(n) \pmod g$ also has period T_g.

For each σ, create $g-1$ correction slots of type σ, so $R_c = w(g-1)$. A correction slot ν consists of a denominator c_ν and a pattern G_ν. If the slot has type σ, then $G_\nu = E_{i_\sigma}$. Choose the c_ν sequentially. For a type-σ slot, impose

$$c_\nu \equiv a_\sigma \pmod{T_g}, \qquad Q_{G_\nu}(c_\nu) > 0, \qquad c_\nu > \max\left\{L, \frac{2R_c}{\delta}\right\},$$

and require $ec_\nu \neq e'c_\mu$ for all previously chosen slots $\mu < \nu$, $e \in \{1\} \cup G_\nu$, $e' \in \{1\} \cup G_\mu$, and $ec_\nu \neq hD_j$ for all $e \in \{1\} \cup G_\nu$, $h \in \{1,2,3,6\}$, and $j \geq J$.

Such choices exist arbitrarily far out in the required arithmetic progression. Positivity holds eventually because Q_{G_ν} has positive leading coefficient. The lower bound and the previously chosen correction denominators exclude only finitely many values. For the main denominators, for fixed e and h, the forbidden values have the form $c_\nu = (h/e)D_j$. Since $D_j \asymp j^2$, only $O(\sqrt{X})$ such forbidden values are at most X, while the arithmetic progression contains $\asymp X$ values at most X. Thus, admissible choices remain.

Define $b_\nu = Q_{G_\nu}(c_\nu)$. Then $b_\nu > 0$. If slot ν has type σ, then, by the congruence condition and the period T_g, $b_\nu \equiv \rho_\sigma \pmod g$. Set

$$C_0 = \sum_{\nu=1}^{R_c} \frac{1}{c_\nu}, \qquad B_* = \sum_{\nu=1}^{R_c} b_\nu.$$

Because $c_\nu > 2R_c/\delta$, we have $C_0 < \delta$.

Each residue class modulo g is represented by a subset-sum of the b_ν. Indeed, since the ρ_σ generate $\mathbb{Z}/g\mathbb{Z}$, any residue r has the form

$$r \equiv \sum_{\sigma=1}^{w} m_\sigma \rho_\sigma \pmod g, \qquad 0 \leq m_\sigma \leq g-1.$$

For each σ, select m_σ of the $g-1$ slots of type σ. Thus, in all cases $g \geq 1$, we have fixed correction slots satisfying $C_0 < \delta$ and representing each residue class modulo g by subset-sums of their p-increments.

Filler denominators. Define $R_0 = \alpha - 1/(Pu_J) - C_0$. Then $R_0 > 0$. Let

$$Y_{\text{corr}} = \begin{cases} 0, & R_c = 0, \\ \max\limits_{1 \leq \nu \leq R_c} \max\limits_{e \in \{1\} \cup G_\nu} ec_\nu, & R_c > 0. \end{cases}$$

Choose an integer Λ such that $\Lambda > \max\{Y_{\text{corr}}, L\}$ and $2^3 \mid \Lambda$. Since $\Lambda R_0 \in \mathbb{Q}_{>0}$, Lemma 3 gives a finite set $F \subseteq \mathbb{N}$ such that

$$\Lambda R_0 = \sum_{f \in F} \frac{1}{f}.$$

Hence,

$$R_0 = \sum_{f \in F} \frac{1}{\Lambda f}.$$

The denominators Λf, $f \in F$, will be fixed filler denominators.

The reciprocal identity and switches. For $N \geq J$, put $\tau_N = Pu_{N+1} = P(P(N+1)+1)$. By telescoping,

$$\sum_{j=J}^{N} \frac{1}{D_j} = \frac{1}{Pu_J} - \frac{1}{Pu_{N+1}} = \frac{1}{Pu_J} - \frac{1}{\tau_N}.$$

Therefore,

$$\sum_{j=J}^{N} \frac{1}{D_j} + \frac{1}{\tau_N} + \sum_{\nu=1}^{R_c} \frac{1}{c_\nu} + \sum_{f \in F} \frac{1}{\Lambda f} = \alpha.$$

Each main denominator D_j may either be left unchanged or replaced by $2D_j$, $3D_j$, and $6D_j$, because

$$\frac{1}{2D_j} + \frac{1}{3D_j} + \frac{1}{6D_j} = \frac{1}{D_j}.$$

This switch changes the p-sum by $A(D_j)$.

Likewise, a correction denominator c_ν may either be left unchanged or replaced by the denominators ec_ν, $e \in G_\nu$, because

$$\sum_{e \in G_\nu} \frac{1}{ec_\nu} = \frac{1}{c_\nu}.$$

This switch changes the p-sum by $b_\nu = Q_{G_\nu}(c_\nu)$. The denominator τ_N and the filler denominators Λf are never switched.

Collision avoidance. We verify that, for all sufficiently large N, all denominators that can occur after any collection of switches are distinct. Since $u_j = 36j+1$ is coprime to 6, each $D_j = u_j u_{j+1}$ is coprime to 6. Thus, for $h = 1, 2, 3, 6$, the valuation pair $(v_2(hD_j), v_3(hD_j))$ is respectively

$$(0,0), \quad (1,0), \quad (0,1), \quad (1,1).$$

If $hD_j = h'D_k$, these valuation pairs force $h = h'$, and then the strict increase of D_j gives $j = k$. Hence, all possible main denominators are distinct.

Also, $\tau_N = 36u_{N+1}$, and u_{N+1} is coprime to 6, so $(v_2(\tau_N), v_3(\tau_N)) = (2, 2)$. This differs from all main valuation pairs, so τ_N never collides with a main denominator.

Within a correction slot, the denominators c_ν and ec_ν, $e \in G_\nu$, are distinct because the multipliers 1 and the elements of G_ν are distinct. Between different correction slots, collisions were explicitly avoided in the construction. Collisions between correction denominators and main denominators were also explicitly avoided.

The filler denominators Λf, $f \in F$, are distinct because the f's are distinct. Since $\Lambda > Y_{\mathrm{corr}}$, all filler denominators are larger than all correction denominators. Since $2^3 \mid \Lambda$, all filler denominators have $v_2(\Lambda f) \geq 3$. On the other hand, $v_2(hD_j) \leq 1$ for $h \in \{1, 2, 3, 6\}$, while $v_2(\tau_N) = 2$. Therefore, no filler denominator collides with a main denominator or with τ_N.

The only remaining possible collisions are between τ_N and fixed correction denominators. Choose N_{coll} so large that $\tau_N > Y_{\mathrm{corr}}$ for $N \geq N_{\mathrm{coll}}$. Then, for $N \geq N_{\mathrm{coll}}$, τ_N does not collide with any correction denominator. Thus, for all $N \geq N_{\mathrm{coll}}$, all denominators that may occur are distinct.

Attainable intervals for fixed N. For $N \geq J$, define the base p-sum

$$B_N = \sum_{j=J}^{N} p(D_j) + p(\tau_N) + \sum_{\nu=1}^{R_c} p(c_\nu) + \sum_{f \in F} p(\Lambda f).$$

After switching a main subset $S \subseteq \{J, \ldots, N\}$ and a correction subset $T \subseteq \{1, \ldots, R_c\}$, the resulting p-sum is

$$B_N + \sum_{j \in S} A(D_j) + \sum_{\nu \in T} b_\nu.$$

Define

$$I_N = [B_N + B_* + M_0,\ B_N + \mu N^{2r}].$$

We show that, for all $N \geq \max(N_{\mathrm{rep}}, N_{\mathrm{coll}})$, all integers $m \in I_N$ are attainable.

Choose a correction subset T such that

$$\Delta_T := \sum_{\nu \in T} b_\nu \equiv m - B_N \pmod{g}.$$

This is possible by the residue-subset property of the correction slots. Since all $b_\nu > 0$, $0 \leq \Delta_T \leq B_*$. Set $M = m - B_N - \Delta_T$. Then $M \equiv 0 \pmod{g}$, while $m \geq B_N + B_* + M_0$ and $\Delta_T \leq B_*$ give $M \geq M_0$, and $m \leq B_N + \mu N^{2r}$ and $\Delta_T \geq 0$ give $M \leq \mu N^{2r}$.

By the finite-window consequence proved above, there is a subset $S \subseteq \{J, \dots, N\}$ such that

$$M = \sum_{j \in S} A(D_j).$$

Switch exactly the main slots in S and the correction slots in T. The reciprocal sum remains α, all denominators are distinct for $N \geq N_{\text{coll}}$, and the resulting p-sum is $B_N + M + \Delta_T = m$. Thus, all integers in I_N are attainable.

Overlap of the intervals. We now show that the intervals I_N cover all sufficiently large integers. The one-step difference is

$$B_{N+1} - B_N = p(D_{N+1}) + p(\tau_{N+1}) - p(\tau_N).$$

Now,

$$p(D_{N+1}) = aP^{2r}N^{2r} + O(N^{2r-1}), \qquad p(\tau_{N+1}) - p(\tau_N) = O(N^{r-1}) = O(N^{2r-1}).$$

Thus,

$$B_{N+1} - B_N = \lambda N^{2r} + O(N^{2r-1}), \qquad \lambda = aP^{2r}.$$

In particular, $B_{N+1} - B_N \geq (\lambda/2)N^{2r}$ eventually, and therefore $B_N \to \infty$.

Let $L_N = B_N + B_* + M_0$ and $U_N = B_N + \mu N^{2r}$, so $I_N = [L_N, U_N]$. Since $\mu > \lambda$, for all sufficiently large N,

$$B_{N+1} - B_N \leq \frac{\mu + \lambda}{2} N^{2r}.$$

Therefore,

$$U_N - L_{N+1} = \mu N^{2r} - (B_{N+1} - B_N) - B_* - M_0 \geq \frac{\mu - \lambda}{2} N^{2r} - B_* - M_0 > 0$$

for all sufficiently large N. Hence, consecutive intervals overlap eventually. Also, $U_N - L_N = \mu N^{2r} - B_* - M_0 > 0$ for all sufficiently large N, and $L_{N+1} - L_N = B_{N+1} - B_N > 0$ eventually.

Choose $N_1 \geq \max\{J, N_{\text{rep}}, N_{\text{coll}}\}$ so large that these conclusions hold for all $N \geq N_1$. Since $L_N \to \infty$, the intervals I_N, $N \geq N_1$, cover all sufficiently large real numbers, and hence all sufficiently large integers.

For each sufficiently large integer m, choose $N \geq N_1$ with $m \in I_N$. The construction above gives a finite set of distinct positive denominators, all greater than L, whose reciprocal sum is α and whose p-sum is m. Sorting these denominators gives $L < n_1 < \cdots < n_k$, with $\sum_{i=1}^k 1/n_i = \alpha$ and $\sum_{i=1}^k p(n_i) = m$. This proves the theorem. □

Corollary 7. *Let $p \in \mathbb{Q}[x]$, and put*

$$A_p = \{p(n) + 1/n : n \in \mathbb{N}\}.$$

If $p = 0$ or p has positive leading coefficient, then A_p is strongly complete: for each finite $B \subseteq A_p$, the finite subset sums from $A_p \setminus B$ contain all sufficiently large integers. If p has negative leading coefficient, then A_p is not strongly complete. Thus, Problem #351 has an affirmative answer after the corrected hypothesis is interpreted as $p = 0$, or $p \neq 0$ with non-negative leading coefficient.

Proof. Write

$$a_n = p(n) + \frac{1}{n}.$$

First suppose $p = 0$. Given a finite set $B \subseteq A_p$, choose L so large that $1/n \notin B$ for all $n > L$. By Lemma 3, each positive integer m is a sum of reciprocals of distinct integers greater than L. Thus, each positive integer is a finite subset sum from $A_p \setminus B$.

Now, suppose p has positive leading coefficient. Choose $D \geq 1$ such that $F(x) := Dp(x) \in \mathbb{Z}[x]$, and put

$$h = \gcd\{F(n) : n \geq 1\}.$$

Then $h \geq 1$. Define $q(x) = F(x)/h$. Since $h \mid F(n)$ for all $n \geq 1$, the polynomial q is integer-valued on all of $\mathbb{Z}$: if $z \in \mathbb{Z}$, choose $n \geq 1$ with $n \equiv z \pmod{h}$, and use $F(n) \equiv F(z) \pmod{h}$. Also, q has positive leading coefficient and no fixed divisor on the positive integers.

Let $B \subseteq A_p$ be finite. For each $b \in B$, the equation $a_n = b$ is equivalent to

$$n(p(n) - b) + 1 = 0,$$

which has only finitely many integer solutions because the polynomial on the left is not identically zero. Also, a_n is eventually injective: if p is nonconstant this follows from

$$a_{n+1} - a_n = p(n+1) - p(n) - \frac{1}{n(n+1)}$$

and the positive leading coefficient of $p(n+1) - p(n)$, while if p is constant it is immediate from $a_n = p + 1/n$. Hence, choose L such that $n > L$ implies $a_n \notin B$, and such that the values a_n, $n > L$, are distinct.

For $r = 1, \ldots, h$, apply Theorem 1 to q, with lower bound L and with $\alpha = r/D$. Taking the maximum of the finitely many resulting thresholds, we find that all sufficiently large integers M are admissible for all these choices of α.

Let m be a sufficiently large integer. Set

$$M = \lceil Dm/h \rceil - 1, \qquad r = Dm - hM.$$

Then $1 \leq r \leq h$, and M is large. By the preceding paragraph, there are distinct integers $n_i > L$ such that

$$\sum_i q(n_i) = M, \qquad \sum_i \frac{1}{n_i} = \frac{r}{D}.$$

Therefore,

$$\sum_i \left(p(n_i) + \frac{1}{n_i} \right) = \frac{h}{D} \sum_i q(n_i) + \frac{r}{D} = \frac{hM + r}{D} = m.$$

The chosen values all lie in $A_p \setminus B$, and are distinct. This proves strong completeness.

Finally, if p has negative leading coefficient, then $a_n < 0$ for all sufficiently large n. Hence, only finitely many elements of A_p are positive, and all finite subset sums from A_p are bounded above by the sum of these positive elements. Thus, A_p cannot be strongly complete. $\square$

Remark 8. The proof uses the rationality of α only when constructing the fixed filler denominators from the positive rational number R_0.

References

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